

8.5 Common Measurements & Conversions

Nominal-versus-actual lumber, unit conversions and the standard dimensions you measure against every day.

Quick reference for common construction measurements, lumber dimensions, and unit conversions. Keep this handy for quick calculations on the job site.

Actual Lumber Dimensions (Nominal vs. Actual)

Nominal Size	Actual Size	Common Uses
1x2	3/4" × 1-1/2"	Trim, furring strips
1x3	3/4" × 2-1/2"	Trim, lattice
1x4	3/4" × 3-1/2"	Trim, boards
1x6	3/4" × 5-1/2"	Trim, fascia, boards
1x8	3/4" × 7-1/4"	Trim, shelving
1x10	3/4" × 9-1/4"	Shelving, trim
1x12	3/4" × 11-1/4"	Shelving, stair treads
2x2	1-1/2" × 1-1/2"	Furring, light framing
2x3	1-1/2" × 2-1/2"	Light framing, blocking
2x4	1-1/2" × 3-1/2"	Wall framing, blocking
2x6	1-1/2" × 5-1/2"	Floor joists, rafters, headers
2x8	1-1/2" × 7-1/4"	Floor joists, rafters, headers
2x10	1-1/2" × 9-1/4"	Floor joists, beams, headers
2x12	1-1/2" × 11-1/4"	Floor joists, beams, headers
4x4	3-1/2" × 3-1/2"	Posts, columns
4x6	3-1/2" × 5-1/2"	Beams, posts
6x6	5-1/2" × 5-1/2"	Large posts, beams

Linear Measurements & Conversions

From	To	Multiply By	Example
Inches	Feet	$\div 12$	36" = 3 feet
Feet	Inches	$\times 12$	5 feet = 60"
Feet	Yards	$\div 3$	9 feet = 3 yards
Yards	Feet	$\times 3$	2 yards = 6 feet
Inches	Centimeters	$\times 2.54$	12" = 30.48 cm
Feet	Meters	$\times 0.3048$	10 feet = 3.048 m

Area & Volume Conversions

Measurement	Conversion	Common Use
1 square foot	144 square inches	Area calculations
1 square yard	9 square feet	Carpet, concrete
1 square (roofing)	100 square feet	Roofing materials
1 cubic foot	1,728 cubic inches	Volume calculations
1 cubic yard	27 cubic feet	Concrete, soil, gravel
1 cubic yard concrete	Covers 81 sq ft at 4" thick	Slab calculations
1 gallon	128 fluid ounces	Paint, liquids
1 gallon	0.1337 cubic feet	Liquid volume

Standard Building Dimensions

Item	Standard Dimension	Notes
Stud spacing	16" or 24" O.C.	On center measurement
Standard ceiling height	8 feet	Residential rooms
Door height (interior)	6'-8" (80")	Standard pre-hung
Door height (exterior)	6'-8" to 8'-0"	Varies by style
Door width (interior)	24", 28", 30", 32", 36"	Common sizes
Door width (exterior)	32", 36"	Most common
Window rough opening	+2" width, +2-1/2" height	Add to window size
Door rough opening	+2" width, +2-1/2" height	Add to door size

Standard Building Dimensions

Item	Standard Dimension	Notes
Countertop height	36"	Kitchen/bathroom standard
Countertop depth	24" to 25"	Standard base cabinet
Upper cabinet height	12", 15", 18", 30", 42"	Common sizes
Stair riser (max)	7-3/4"	Residential code
Stair tread (min)	10"	Residential code

Quick Tips

- When measuring, always double-check and measure twice before cutting
- "O.C." means "on center" — measured from center of one member to center of the next
- Rough openings are larger than actual door/window sizes to allow for shimming and adjustment
- Always verify local code requirements as they may differ from these standards